Ho Lok Cheung

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Education

University of California, Los Angeles Los Angeles, CA | *Expected July 2027*

Bachelor of Science in Computer Science and Linguistics

Relevant Coursework: Algorithms and Complexity, Software Construction, Computer Organization, Programming Languages, Artificial Intelligence, Computational Linguistics

Experience

Student Software Developer Aug 2026 – Present

UCLA Digital and Technology Solutions, Digital Spaces *Los Angeles, CA*

- Improve reliability of classroom applications supporting UCLA's Learning Spaces Modernization Project by debugging issues and contributing fixes to shared codebases.
- Rewrite existing code in these classroom apps alongside full time engineers on the Digital Spaces team, documenting each change so the team can maintain it.

Project Manager Sep 2024 – Dec 2024

NASA L'SPACE Academy *Tempe, AZ*

- Led a team of 20 across science, engineering, and program roles; authored the full mission design review for a semiautonomous lunar rover.

Research Intern | **Paper** Jun 2024 – Aug 2024

Stanford University *Palo Alto, CA*

- Built an evolutionary simulator in Python on a spatial grid, modeling mutation, genome recombination, and natural selection under configurable mutation rates and spatial constraints.
- Structured the system around modular Creature, State, and Action classes with a live visualization canvas.
- First author of the resulting paper, which showed populations readapting after the survival zone was relocated.

Operations Intern Jun 2023 – Aug 2023

Hang Seng Bank *Hong Kong*

- Wrote SQL queries to extract trade data and built Excel workflows that cleaned and tracked daily trade records.

Projects

TouchSpace | **Code** Oct 2026

Claude for Public Benefit Prize (Anthropic), Cal Hacks 13.0 *Python, FastAPI, React, TypeScript, Claude API*

- Built an accessibility platform on a three person team that turns LiDAR iPhone scans into tactile maps on a 2,400 pin Dot Pad display and spoken answers for blind and visually impaired users.
- Built a FastAPI backend with 20 API endpoints and a React web app; the backend suite contains 153 automated test functions. Integrated Apple RoomPlan to convert scans into floor plans and walkable navigation graphs.
- Implemented Dijkstra route planning and a Claude powered assistant that interprets spoken questions but computes every route from map data, so the AI cannot invent a path.

Naccent | **Code** Jan 2025

Best Student Life Hack, SB Hacks XI *Next.js, Python, Flask, Whisper API*

- Built an AI pronunciation trainer that transcribes speech with the OpenAI Whisper API and scores it phoneme by phoneme through SpeechAce and a Flask service, returning per sound scores and native speaker comparison.
- Shipped the Next.js frontend with Chart.js progress analytics after replacing an initial scoring approach built only on an LLM, which handled accents and phonetic detail poorly.

GreenMate | **Demo** Oct 2024

1st Place, Columbia University Hackathon *Snap Spectacles, Lens Studio, AR*

- Built an AR app for Snap Spectacles in Lens Studio that overlays live plant health data in the wearer's view.

Technical Skills

Languages: C++, C, Python, JavaScript, TypeScript, SQL, x86 Assembly

Web and Backend: React, Next.js, Node.js, Express.js, FastAPI, Flask, MongoDB, REST APIs, JWT Authentication

Tools and AI: Git, GitHub, Linux, Bash, SSH, Regex, OpenAI Whisper API, Claude API, Claude Code

Concepts: Data Structures and Algorithms, Multithreading (pthreads), Object Oriented Design

Spoken Languages: Cantonese (native), Mandarin (fluent), English (fluent)

Honors and Activities

- **Regents Scholar**, UCLA's top 2% merit scholarship | **Dean's Honors List**, Spring 2026
- **USA Team Handball Academic All-American**, 1st Team 2026 | **NCAA Division I Rugby**, UCLA